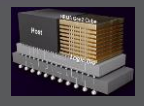
2025 Summer Intern Presentation

Intern: Yoshiki Takeuchi

Department: APTD WLT Thinning

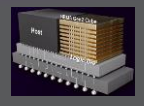
Agenda



- Self Introduction
- Internship Experience Sharing
- Background
- Design of Experiment
- Experiment Process & Data Collection
- Data analysis
  - Wafer Grouping using Correlation Matrix
  - Wafer Grouping
  - Tool Alarm: Software Issue
- Results
- Conclusion and Future work
- Reference

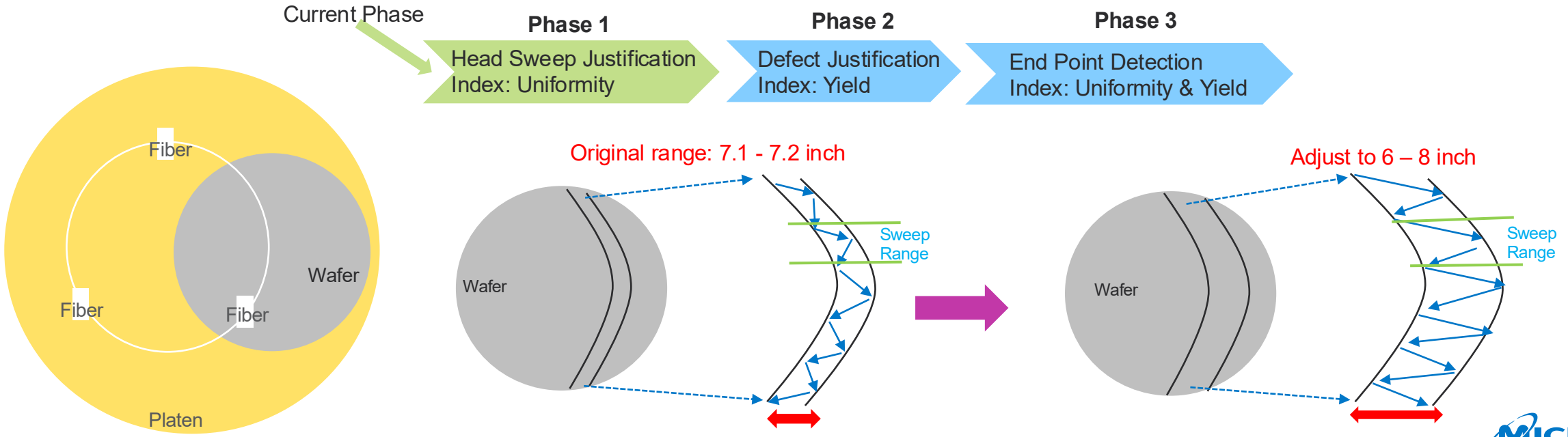
Intern Project:

Investigate CMP Polishing Head Sweep Effect



Background

- **Objective:** Investigate the OX CMP polish head sweep range to determine its potential benefits on improving wafer uniformity and thickness.
- **Methods:**
  - In this CMP experiment, several parameters can be adjusted. There are various methods that can affect CMP uniformity, such as head sweep range, slurry arm position, and dresser sweep. We decided to start with the most intuitive parameter: the **polishing head sweep.**
  - Adjust CMP sweep range from 6 inch – 8 inch with 5 DOE with 25 blanket wafer.
- **Hypothesis:**
  1. When the head is moving, it helps the wafer surface to have **fresh slurry.**
     - It can decrease by-product to build and prevent wafer to staying from same place for too long → Improve defect performance and planarization efficiency.
  2. Increasing the head sweep range allows the system to collect endpoint data over a **wider range** and adjust zone pressure more effectively.
     - Increase the detection area on the wafer → earlier stage to assess its condition.
     - **NO endpoint data** for this experiment. (Due to it is Blanket wafer and we want to focus on profile tuning)



Liu et al., *ECS J. Solid State Sci. Technol.*, 2024. https://doi.org/10.1149/2162-8777/ad38d8

Thank you so much!
Any Questions?